UNIVERSIDAD DE GUAYAQUIL

FACULTAD DE CIENCIAS, MATEMATICAS Y FISICAS

CARRERA DE SOFTWARE

MATERIA:

GESTION DE LA CONFIGURACION DEL SOFTWARE

TEMA DE ACTIVIDAD:

CONTROL DE CAMBIOS GIT + MIGRACION BD

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CICLO ESCOLAR:

2025 – 2026 CICLO II

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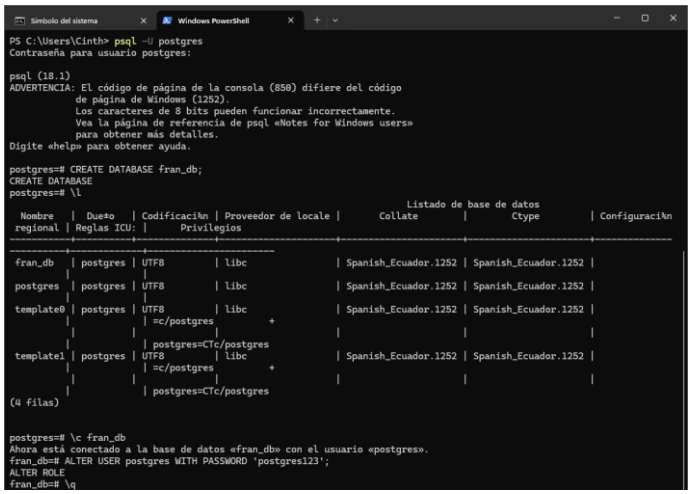
CONTROL DE CAMBIO GIT + MIGRACION BD

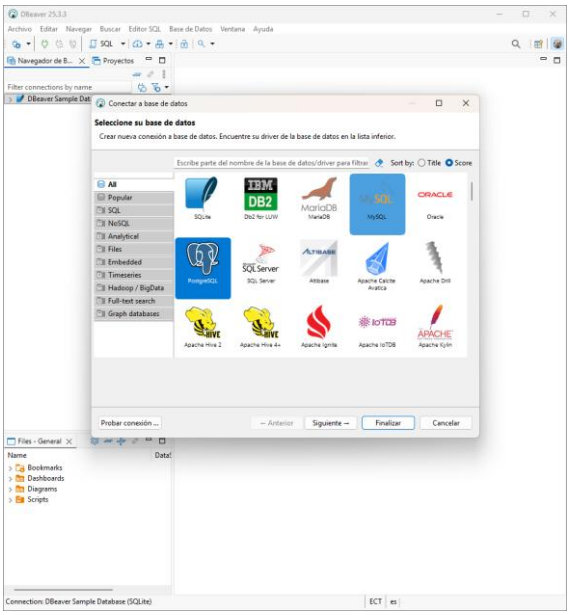
1. Requisitos Mínimos

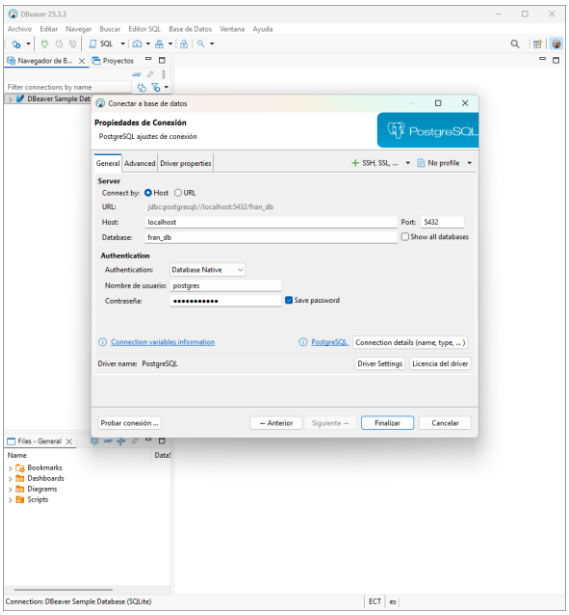
- Git instalado.
- Python 3.10+ instalado.
- VS Code (recomendado).
- DBeaver Community instalado (Windows o Linux).
- Una BD disponible: PostgreSQL recomendada (puede estar en Docker/Portainer).

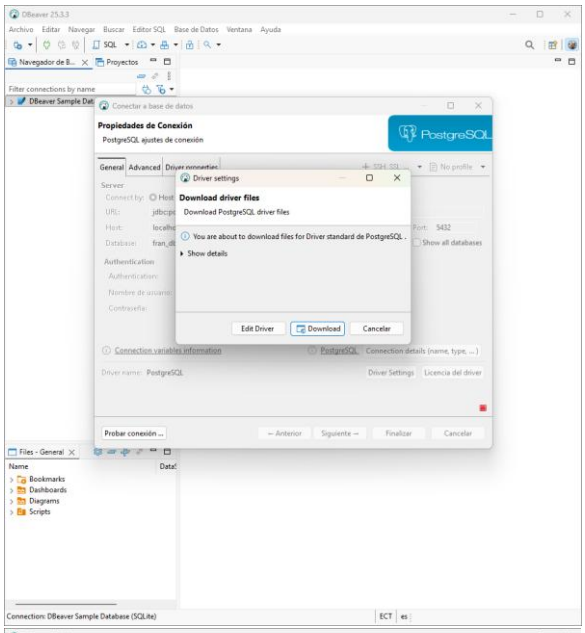
2. Base de Datos Estándar del Laboratorio

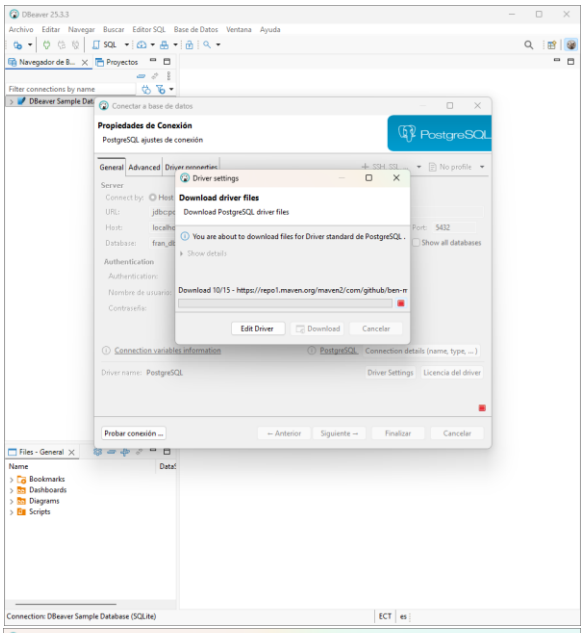


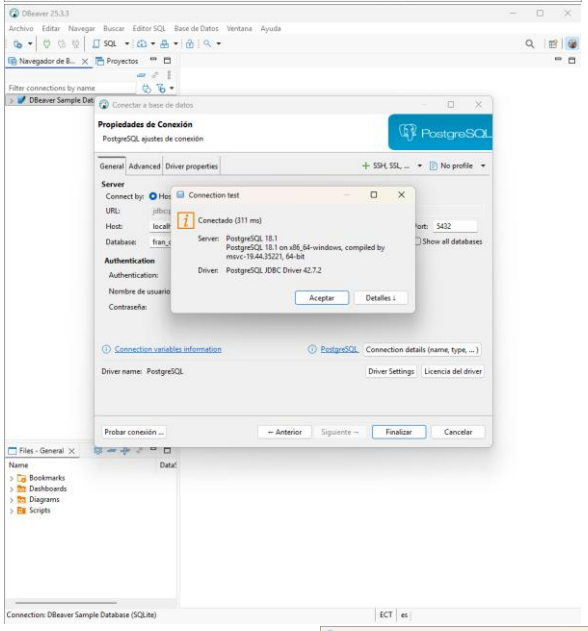
3. Conexión y Ejecución SQL con DBeaver

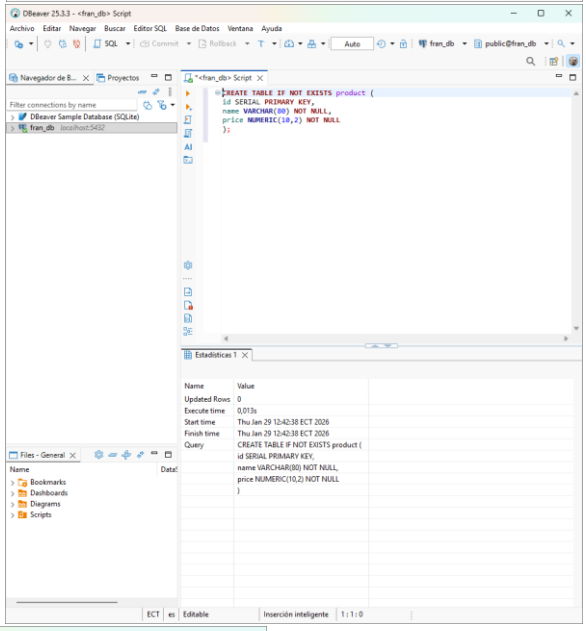


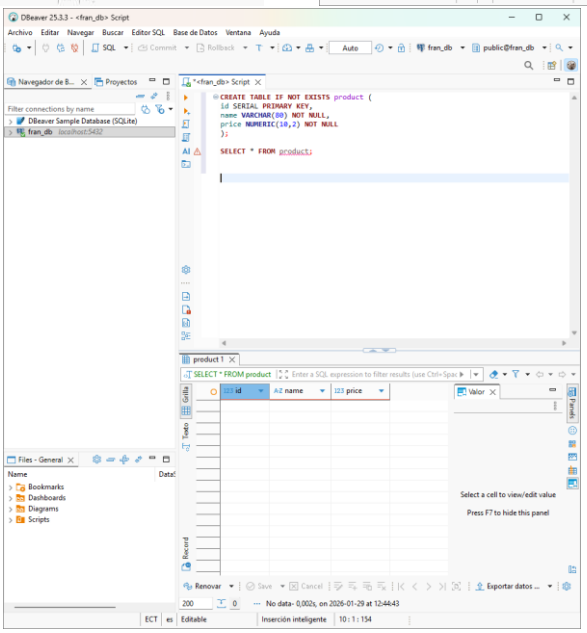



Propiedades de Conexión
PostgreSQL ajustes de conexión
Driver settings
Download driver files
Download PostgreSQL driver files
You are about to download files for Driver standard de PostgreSQL .
Show details
Edit Driver
Download
Cancel


Propiedades de Conexión
PostgreSQL ajustes de conexión
Driver settings
Download driver files
Download PostgreSQL driver files
You are about to download files for Driver standard de PostgreSQL .
Download 10/15 - https://repo1.maven.org/maven2/com/github/ben-m
Edit Driver
Download
Cancel

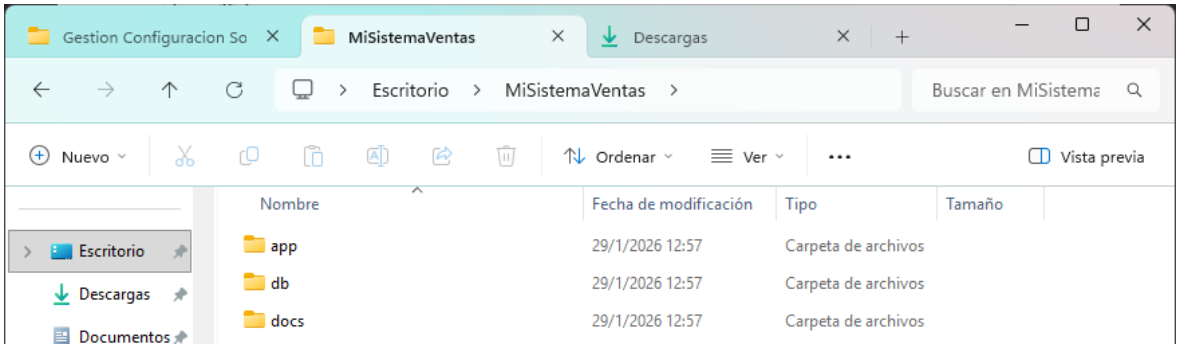

Propiedades de Conexión
PostgreSQL ajustes de conexión
Connection test
Conectado (311 ms)
Server: PostgreSQL 18.1
PostgreSQL 18.1 on x86_64-windows, compiled by msvc-19.44.35221, 64-bit
Driver: PostgreSQL JDBC Driver 42.7.2
Aceptar
Detalles
Finalizar
Cancelar


DBeaver 25.3.3 - <fran_db> Script
CREATE TABLE IF NOT EXISTS product (
id SERIAL PRIMARY KEY,
name VARCHAR(80) NOT NULL,
price NUMERIC(10,2) NOT NULL
);
Estadísticas 1
Updated Rows 0
Execute time 0,013s
Start time Thu Jan 29 12:42:38 ECT 2026
Finish time Thu Jan 29 12:42:38 ECT 2026
Query CREATE TABLE IF NOT EXISTS product (
id SERIAL PRIMARY KEY,
name VARCHAR(80) NOT NULL,
price NUMERIC(10,2) NOT NULL
)

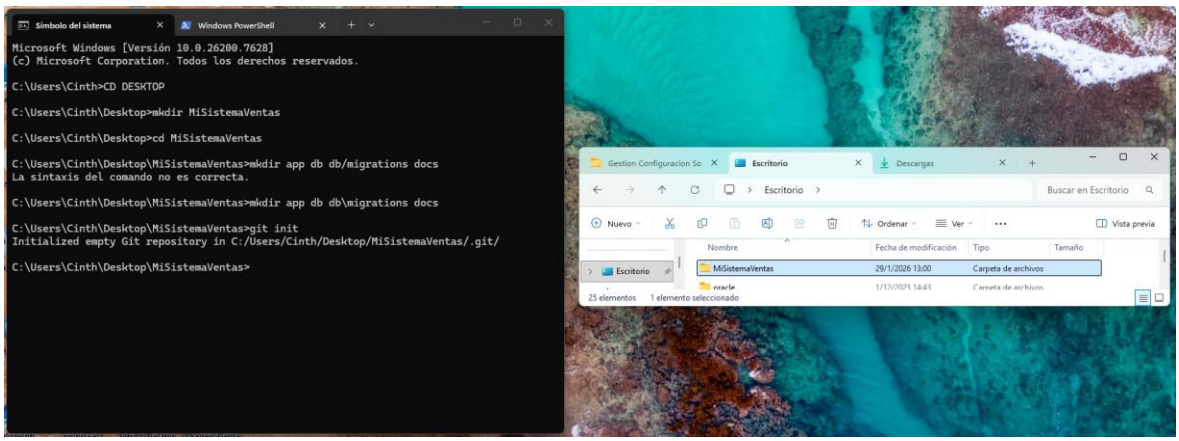

DBeaver 25.3.3 - <fran_db> Script
CREATE TABLE IF NOT EXISTS product (
id SERIAL PRIMARY KEY,
name VARCHAR(80) NOT NULL,
price NUMERIC(10,2) NOT NULL
);
SELECT * FROM product;
product 1
SELECT * FROM product
id
name
price
Select a cell to view/edit value
Press F7 to hide this panel

4. Proyecto API + Git

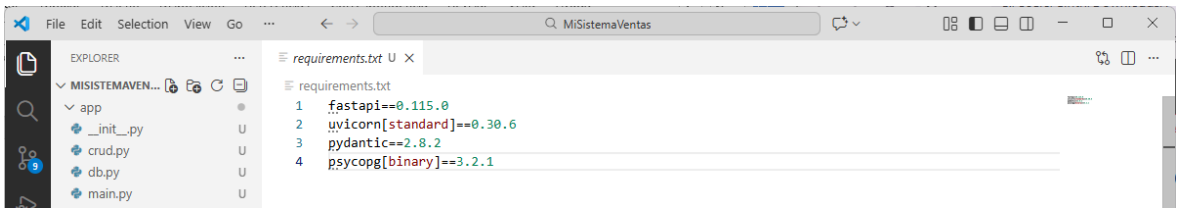
4.1.Estructura



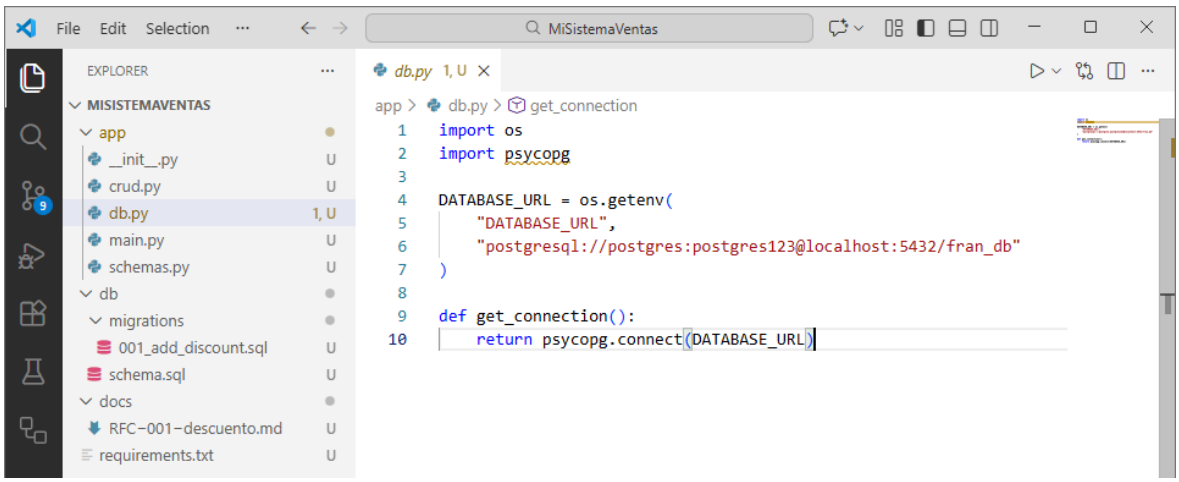
4.2.Crear proyecto e iniciar Git



4.3. Requirements.txt



4.4.Código Base



```python
from pydantic import BaseModel, Field

class ProductCreate(BaseModel):
    name: str = Field(min_length=2, max_length=80)
    price: float = Field(gt=0)

class ProductOut(BaseModel):
    id: int
    name: str
    price: float
```

```python
from app.db import get_connection

def create_product(name: str, price: float) -> dict:
    with get_connection() as conn:
        with conn.cursor() as cur:
            cur.execute(
                """
                INSERT INTO product(name, price)
                VALUES (%s, %s)
                RETURNING id, name, price
                """,
                (name, price)
            )
            row = cur.fetchone()
            conn.commit()
            return {"id": row[0], "name": row[1], "price": float(row[2])}

def list_products() -> list[dict]:
    with get_connection() as conn:
        with conn.cursor() as cur:
            cur.execute("SELECT id, name, price FROM product ORDER BY id")
            rows = cur.fetchall()
            return [{"id": r[0], "name": r[1], "price": float(r[2])} for r in rows]
```

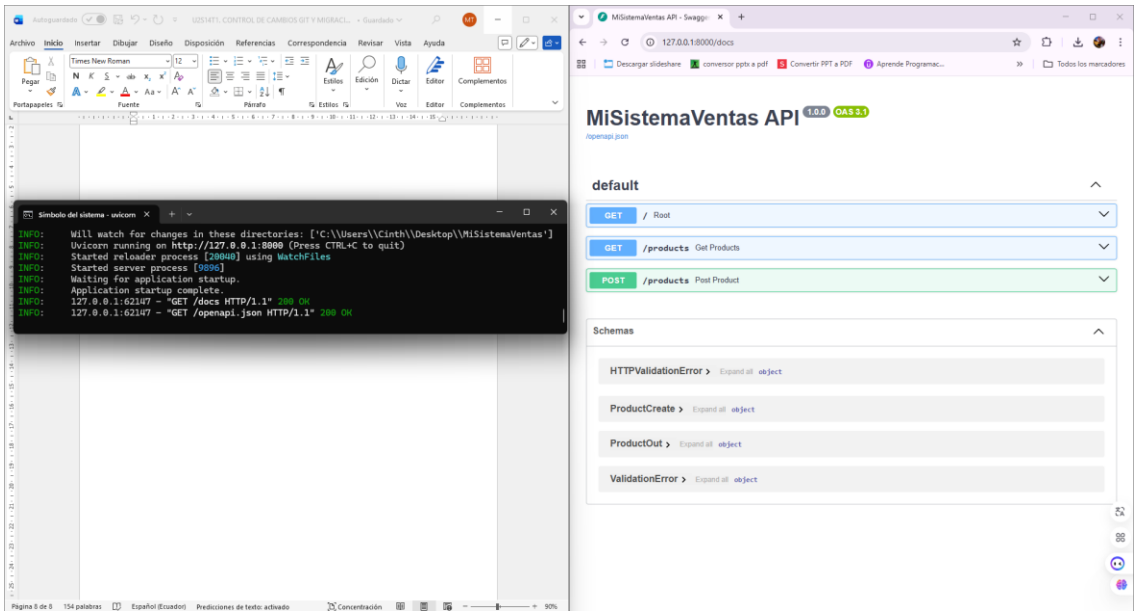
```python
from fastapi import FastAPI
from app.schemas import ProductCreate, ProductOut
from app import crud

app = FastAPI(title="MiSistemaVentas API", version="1.0.0")

@app.get("/")
def root():
    return {"message": "API OK. Ve a /docs o /products"}

@app.get("/products", response_model=list[ProductOut])
def get_products():
    return crud.list_products()

@app.post("/products", response_model=ProductOut)
def post_product(payload: ProductCreate):
    return crud.create_product(payload.name, payload.price)
```

4.5.Crear Entorno Virtual

```
Símbolo del sistema
Microsoft Windows [Versión 10.0.26200.7628]
(c) Microsoft Corporation. Todos los derechos reservados.

C:\Users\Cinth>cd C:\Users\Cinth\Desktop\MiSistemaVentas

C:\Users\Cinth\Desktop\MiSistemaVentas>.\.venv\Scripts\Activate.ps1

C:\Users\Cinth\Desktop\MiSistemaVentas>python --version
Python 3.14.2

C:\Users\Cinth\Desktop\MiSistemaVentas>.venv\Scripts\activate.bat

(.venv) C:\Users\Cinth\Desktop\MiSistemaVentas>python --version
Python 3.10.0

(.venv) C:\Users\Cinth\Desktop\MiSistemaVentas>pip install -r requirements.txt
Collecting fastapi==0.115.0
  Downloading fastapi-0.115.0-py3-none-any.whl (94 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 94 kB 129 kB/s
Collecting uvicorn[standard]==0.30.6
  Downloading uvicorn-0.30.6-py3-none-any.whl (62 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 62 kB 51 kB/s
Collecting pydantic==2.8.2
  Downloading pydantic-2.8.2-py3-none-any.whl (423 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 423 kB 2.2 MB/s
Collecting psycopg2-binary==2.9.9
  Downloading psycopg2_binary-2.9.9-cp310-cp310-win_amd64.whl (1.2 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.2 MB 3.3 MB/s
Collecting starlette<0.39.0,>=0.37.2
  Downloading starlette-0.38.6-py3-none-any.whl (71 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 71 kB 498 kB/s
Collecting typing-extensions>=4.8.0
  Downloading typing_extensions-4.15.0-py3-none-any.whl (44 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 44 kB 218 kB/s
Collecting h11>=0.8
  Downloading h11-0.16.0-py3-none-any.whl (37 kB)
Collecting click>=7.0
  Downloading click-8.3.1-py3-none-any.whl (108 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 108 kB 6.4 MB/s
Collecting watchfiles>=0.13
  Downloading watchfiles-1.1.1-cp310-cp310-win_amd64.whl (287 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 287 kB 6.4 MB/s
```

```
Símbolo del sistema
Collecting httptools>=0.5.0
  Downloading httptools-0.7.1-cp310-cp310-win_amd64.whl (86 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 86 kB 259 kB/s
Collecting python-dotenv>=0.13
  Downloading python_dotenv-1.2.1-py3-none-any.whl (21 kB)
Collecting websockets>=10.4
  Downloading websockets-16.0-cp310-cp310-win_amd64.whl (178 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 178 kB 3.3 MB/s
Collecting pyyaml>=5.1
  Downloading pyyaml-6.0.3-cp310-cp310-win_amd64.whl (158 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 158 kB 3.2 MB/s
Collecting colorama>=0.4
  Downloading colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Collecting pydantic-core==2.20.1
  Downloading pydantic_core-2.20.1-cp310-none-win_amd64.whl (1.9 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.9 MB 3.3 MB/s
Collecting annotated-types>=0.4.0
  Downloading annotated_types-0.7.0-py3-none-any.whl (13 kB)
Collecting anyio<5,>=3.4.0
  Downloading anyio-4.12.1-py3-none-any.whl (113 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 113 kB 3.2 MB/s
Collecting idna>=2.8
  Downloading idna-3.11-py3-none-any.whl (71 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 71 kB 869 kB/s
Collecting exceptiongroup>=1.0.2
  Downloading exceptiongroup-1.3.1-py3-none-any.whl (16 kB)
Installing collected packages: typing-extensions, idna, exceptiongroup, colorama, pydantic-core, h11, click, anyio, anno
tated-types, websockets, watchfiles, uvicorn, starlette, pyyaml, python-dotenv, pydantic, httptools, psycopg2-binary, fa
stapi
Successfully installed annotated-types-0.7.0 anyio-4.12.1 click-8.3.1 colorama-0.4.6 exceptiongroup-1.3.1 fastapi-0.115.
0 h11-0.16.0 httptools-0.7.1 idna-3.11 psycopg2-binary-2.9.9 pydantic-2.8.2 pydantic-core-2.20.1 python-dotenv-1.2.1 pyy
aml-6.0.3 starlette-0.38.6 typing-extensions-4.15.0 uvicorn-0.30.6 watchfiles-1.1.1 websockets-16.0
```

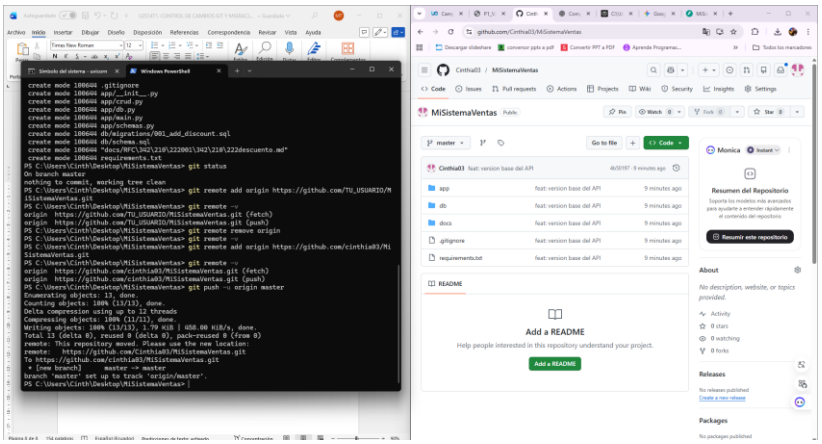
4.6.Levantar Entorno Virtual



5. Control de Versiones

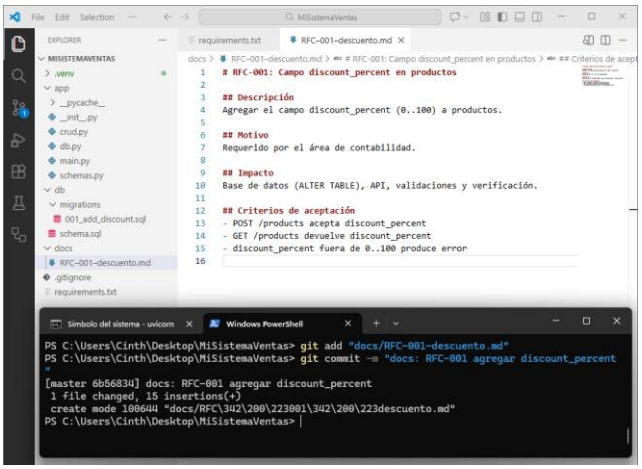
```
Símbolo del sistema - uvicorn    Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. Todos los derechos reservados.

Instale la versión más reciente de PowerShell para obtener nuevas características y mejoras. htt
ps://aka.ms/PSWindows

PS C:\Users\Cinth> cd C:\Users\Cinth\Desktop\MiSistemaVentas
PS C:\Users\Cinth\Desktop\MiSistemaVentas> @"
>> # Entorno virtual
>> .venv/
>> venv/
>>
>> # Cache de Python
>> __pycache__/
>> *.pyc
>>
>> # Variables de entorno
>> .env
>>
>> # IDE / Editor
>> .vscode/
>> .idea/
>> "@ | Out-File -Encoding utf8 .gitignore
PS C:\Users\Cinth\Desktop\MiSistemaVentas> dir .gitignore


    Directorio: C:\Users\Cinth\Desktop\MiSistemaVentas


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----         29/1/2026     15:32            134 .gitignore
```

```
Símbolo del sistema - uvicorn    Windows PowerShell

PS C:\Users\Cinth\Desktop\MiSistemaVentas> git init
Reinitialized existing Git repository in C:/Users/Cinth/Desktop/MiSistemaVentas/.git/
PS C:\Users\Cinth\Desktop\MiSistemaVentas> git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore
        app/
        db/
        docs/
        requirements.txt

nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\Cinth\Desktop\MiSistemaVentas> git add .
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touc
hes it
PS C:\Users\Cinth\Desktop\MiSistemaVentas> git commit -m "feat: version base del API"
[master (root-commit) 4b50197] feat: version base del API
 10 files changed, 78 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 app/__init__.py
 create mode 100644 app/crud.py
 create mode 100644 app/db.py
 create mode 100644 app/main.py
 create mode 100644 app/schemas.py
 create mode 100644 db/migrations/001_add_discount.sql
 create mode 100644 db/schema.sql
 create mode 100644 "docs/RFC\342\210\222001\342\210\222descuento.md"
 create mode 100644 requirements.txt
PS C:\Users\Cinth\Desktop\MiSistemaVentas> git status
On branch master
nothing to commit, working tree clean
PS C:\Users\Cinth\Desktop\MiSistemaVentas>
```

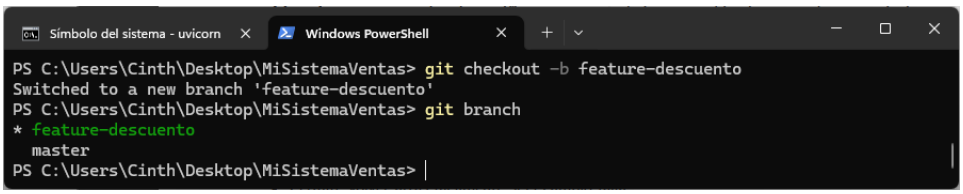


6. Control de Cambios

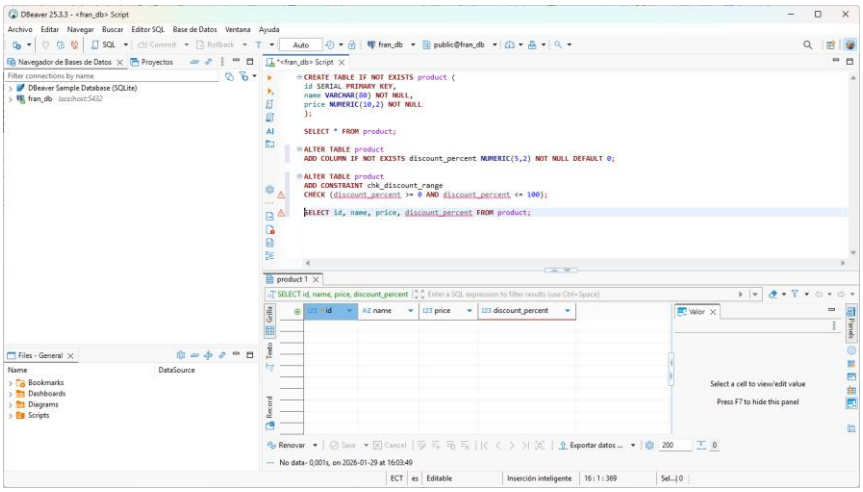
6.1.RFC



6.2.Crear rama



6.3.Migración SQL



6.4.Actualizar API

```python
from pydantic import BaseModel, Field

class ProductCreate(BaseModel):
    name: str = Field(min_length=2, max_length=80)
    price: float = Field(gt=0)
    discount_percent: float = Field(default=0, ge=0, le=100)

class ProductOut(BaseModel):
    id: int
    name: str
    price: float
    discount_percent: float
```

```python
from app.db import get_connection

def create_product(name: str, price: float, discount_percent: float = 0) -> dict:
    with get_connection() as conn:
        with conn.cursor() as cur:
            cur.execute("""
                INSERT INTO product(name, price, discount_percent)
                VALUES (%s, %s, %s)
                RETURNING id, name, price, discount_percent
            """, (name, price, discount_percent))
            row = cur.fetchone()
        conn.commit()
    return {
        "id": row[0],
        "name": row[1],
        "price": float(row[2]),
        "discount_percent": float(row[3])
    }

def list_products() -> list[dict]:
    with get_connection() as conn:
        with conn.cursor() as cur:
            cur.execute("SELECT id, name, price, discount_percent FROM product ORDER BY id")
            rows = cur.fetchall()
    return [
        {
            "id": r[0],
            "name": r[1],
            "price": float(r[2]),
            "discount_percent": float(r[3])
        }
        for r in rows
    ]
```

```python
from fastapi import FastAPI
from app.schemas import ProductCreate, ProductOut
from app import crud

app = FastAPI(title="MiSistemaVentas API", version="1.1.0")

@app.get("/products", response_model=list[ProductOut])
def get_products():
    return crud.list_products()

@app.post("/products", response_model=ProductOut)
def post_product(payload: ProductCreate):
    return crud.create_product(
        payload.name,
        payload.price,
        payload.discount_percent
    )
```

6.5.QA

```
Microsoft Windows [Versión 10.0.26200.7628]
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C:\Users\Cinth>cd C:\Users\Cinth\Desktop\MiSistemaVentas

C:\Users\Cinth\Desktop\MiSistemaVentas>curl http://127.0.0.1:8000/products
[]
C:\Users\Cinth\Desktop\MiSistemaVentas>curl -X POST http://127.0.0.1:8000/products -H "Content-T
ype: application/json" -d "{\"name\":\"Monitor\",\"price\":180,\"discount_percent\":10}"
{"id":1,"name":"Monitor","price":180.0,"discount_percent":10.0}
C:\Users\Cinth\Desktop\MiSistemaVentas>curl -X POST http://127.0.0.1:8000/products -H "Content-T
ype: application/json" -d "{\"name\":\"Laptop\",\"price\":900,\"discount_percent\":150}"
{"detail":[{"type":"less_than_equal","loc":["body","discount_percent"],"msg":"Input should be le
ss than or equal to 100","input":150,"ctx":{"le":100.0}}]}
```

6.6. Commit del cambio

```
C:\Users\Cinth\Desktop\MiSistemaVentas>git add .

C:\Users\Cinth\Desktop\MiSistemaVentas>git commit -m "feat: agregar discount_percent (migración BD + A
PI + validaciones)"
[feature-descuento b98f225] feat: agregar discount_percent (migración BD + API + validaciones)
 6 files changed, 51 insertions(+), 19 deletions(-)
 delete mode 100644 "docs/RFC\342\210\222001\342\210\222descuento.md"

C:\Users\Cinth\Desktop\MiSistemaVentas>git branch
* feature-descuento
  master

C:\Users\Cinth\Desktop\MiSistemaVentas>git diff

C:\Users\Cinth\Desktop\MiSistemaVentas>git log --oneline --graph --decorate --all
* b98f225 (HEAD -> feature-descuento) feat: agregar discount_percent (migración BD + API + validacione
s)
* 6b56834 (master) docs: RFC-001 agregar discount_percent
* 4b50197 (origin/master) feat: version base del API
```

```
C:\Users\Cinth\Desktop\MiSistemaVentas>git show b98f2255
commit b98f22554b8a289e9f3ac43d858833d1c0eefbac (HEAD -> feature-descuento)
Author: Cinthia Tenesaca <cinthia2003maria@gmail.com>
Date:   Thu Jan 29 16:18:15 2026 -0500

    feat: agregar discount_percent (migración BD + API + validaciones)

diff --git a/app/crud.py b/app/crud.py
index 627c3c2..0725145 100644
--- a/app/crud.py
+++ b/app/crud.py
@@ -1,23 +1,33 @@
 from app.db import get_connection

-def create_product(name: str, price: float) -> dict:
+def create_product(name: str, price: float, discount_percent: float = 0) -> dict:
     with get_connection() as conn:
         with conn.cursor() as cur:
-            cur.execute(
-                """
-                INSERT INTO product(name, price)
-                VALUES (%s, %s)
-                RETURNING id, name, price
-                """,
-                (name, price)
-            )
+            cur.execute("""
+                INSERT INTO product(name, price, discount_percent)
+                VALUES (%s, %s, %s)
+                RETURNING id, name, price, discount_percent
+            """, (name, price, discount_percent))
             row = cur.fetchone()
-            conn.commit()
-            return {"id": row[0], "name": row[1], "price": float(row[2])}

C:\Users\Cinth\Desktop\MiSistemaVentas>git blame app/crud.py
^4b50197 (Cinthia Tenesaca 2026-01-29 15:34:08 -0500  1) from app.db import get_connection
^4b50197 (Cinthia Tenesaca 2026-01-29 15:34:08 -0500  2)
b98f2255 (Cinthia Tenesaca 2026-01-29 16:18:15 -0500  3) def create_product(name: str, price: float, d
iscount_percent: float = 0) -> dict:
^4b50197 (Cinthia Tenesaca 2026-01-29 15:34:08 -0500  4)     with get_connection() as conn:
^4b50197 (Cinthia Tenesaca 2026-01-29 15:34:08 -0500  5)         with conn.cursor() as cur:
b98f2255 (Cinthia Tenesaca 2026-01-29 16:18:15 -0500  6)             cur.execute("""
b98f2255 (Cinthia Tenesaca 2026-01-29 16:18:15 -0500  7)                 INSERT INTO product(name, pri
ce, discount_percent)
b98f2255 (Cinthia Tenesaca 2026-01-29 16:18:15 -0500  8)                 VALUES (%s, %s, %s)
b98f2255 (Cinthia Tenesaca 2026-01-29 16:18:15 -0500  9)                 RETURNING id, name, price, di
scount_percent
b98f2255 (Cinthia Tenesaca 2026-01-29 16:18:15 -0500 10)             """, (name, price, discount_perce
nt))
^4b50197 (Cinthia Tenesaca 2026-01-29 15:34:08 -0500 11)             row = cur.fetchone()
```

6.7. Merge a master

```
C:\Users\Cinth\Desktop\MiSistemaVentas>git checkout master
Switched to branch 'master'
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

C:\Users\Cinth\Desktop\MiSistemaVentas>git merge feature-descuento
Updating 6b56834..b98f225
Fast-forward
 app/crud.py                                        | 36 +++++++++++++++--------
 app/main.py                                        | 13 ++++----
 app/schemas.py                                     |  2 ++
 db/migrations/001_add_discount.sql                 |  8 +++++
 db/schema.sql                                      | 11 +++++++
 "docs/RFC\342\210\222001\342\210\222descuento.md" |  0
 6 files changed, 51 insertions(+), 19 deletions(-)
 delete mode 100644 "docs/RFC\342\210\222001\342\210\222descuento.md"
```